

Parchi

A prescription reader that catches the medicine you are already taking.



All India Hackathon 2026 · Axcentra

Track: Healthcare and Digital Public Good

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PROBLEM STATEMENT

Two doctors, two paper slips, one molecule taken twice.

A patient visits a doctor for fever and gets Dolo 650. Ten days later a different doctor, who cannot see the first slip, writes Calpol 650 for body pain. Both are paracetamol. The patient thinks they are two different medicines and takes both.

This is not a rare accident. It is the normal outcome of a system where the medicine history lives on loose paper:

- Most OPD prescriptions here are still handwritten, and the pharmacy counter reads them in well under a minute.
- One molecule is sold under dozens of brand names. Nothing on the strip tells an ordinary patient that two brands are the same drug.
- Patients above 60 are often on five or more medicines from three or more prescribers. No single person in that chain holds the full list.

The information needed to prevent this is already printed on the paper. It just never reaches anything capable of checking it.

What goes wrong

Paracetamol crossing 4 g a day is toxic to the liver. Two painkillers from the same family stack their stomach and kidney risk. A sedative added on top of another one is how elderly patients fall at night.

Why existing apps do not solve it

Drug interaction checkers need internet and expect you to type correct generic names. Hospital software is built for pharmacists inside hospitals. Neither helps the person standing outside a chemist shop with a folded paper slip and no signal.

PROPOSED SOLUTION

Photograph the slip. Parchi reads it, remembers it, and warns you before you swallow.

1 Read

On-device OCR pulls out brand names, strengths and timings. Handwriting is messy, so the user confirms the extracted list in two taps instead of trusting the machine blindly.

2 Resolve

Every brand is mapped to its actual composition using an offline brand-to-molecule index built from public drug listings. This mapping is the hard part, and it is what makes the rest possible.

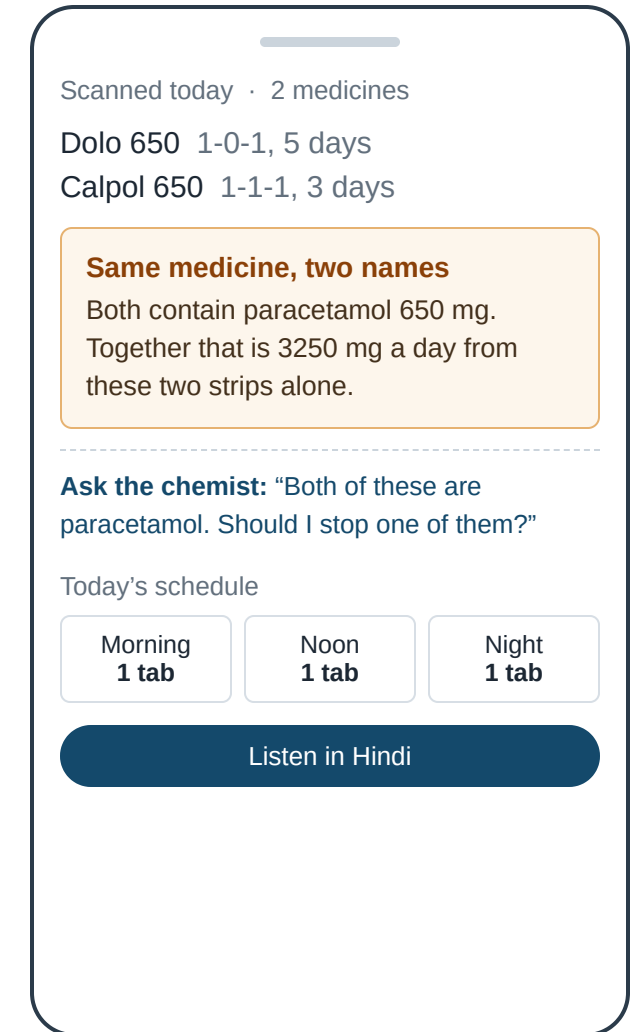
3 Check

Rules run across every prescription the phone has ever scanned, not just today's: repeated molecule, daily dose crossing its ceiling, two drugs of the same class, risky pairs, age and pregnancy flags.

4 Explain

A spoken schedule in the patient's own language plus a printable colour card, so it works for someone who cannot read the slip at all.

Three rules we set for ourselves: nothing leaves the phone, everything works with the network off, and Parchi never tells anyone to change a dose. It hands the patient the exact question to ask the pharmacist.



Built to run on a cheap Android phone with the data switched off.

App	Flutter, Android first
Text extraction	ML Kit on-device OCR, with a small handwriting recognition model exported to ONNX for drug names
Brand matching	Offline index with fuzzy and phonetic matching, so “DOLO650”, “dolo 650” and a half-wrong OCR read all land on the same molecule
Safety rules	SQLite store of molecules, dose ceilings, class groups and interaction pairs, with the rule engine in Dart
Voice	On-device text to speech, Hindi and English at first
Modelling	Wolfram Language to build the dose ceiling tables and to sanity check the interaction graph before it ships
Updates only	FastAPI and Postgres, used purely to push new drug database versions. The app never depends on it to work.

What will exist by the end of the hackathon

- An installable APK that runs fully offline
- Around 1,500 Indian brand names mapped to their molecules
- Five live checks: repeated molecule, dose ceiling, same class stacking, risky pair, age or pregnancy flag
- Spoken schedule in Hindi and English
- Accuracy measured on roughly 50 real prescriptions we collect ourselves

We are deliberately not building a login, a cloud sync or a chatbot. None of them make the warning arrive faster.

One warning, delivered at the counter, before the strip is opened.

60+

Our first users: patients on five or more medicines from more than one doctor

0

Bytes of prescription data leaving the phone

0 MB

Internet needed for a check to run

How we will know it worked. We take pairs of real prescriptions written by different doctors for the same person and measure how many duplicate molecules Parchi catches, and how many it misses. That number is the whole product.

Where it goes after the hackathon. The same scan works from the other side of the counter, so chemists and Jan Aushadhi stores become the first distribution route. ASHA workers already carry the elderly patients who need it most. Linking to ABHA later turns a folder of scans into a real medicine history.

What could break this

Bad handwriting will defeat the OCR sometimes. We answer that by making the user confirm the list before any check runs, so a wrong read never turns into a wrong warning.

The brand database will never be complete. Unknown brands are shown as unknown rather than as safe, and users can correct a mapping in one tap so the next version is better.

Nobody is going to hand every patient a pharmacist. A phone camera and a good offline database get surprisingly close.